IE 425 Data Mining · Fall 2026 · Term project checkpoint

Forecasting daily bike arrivals in Oslo Group 11 · 6 November 2026

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1 How we frame the problem

Unit of analysis. One *cluster-day*: a pair (i, d) of a station cluster $i \in \{1, \ldots, 5\}$ (Center, East, North, North-west, West) and an Oslo calendar day d (00:00–23:59, Europe/Oslo). Each target day contributes five rows.

Target. The inflow of cluster i on day d,

$$y_{i,d} \;=\; \sum_{s \in \mathcal{S}_i} a_{s,d},$$

where $\mathcal{S}_i$ is the set of scored stations assigned to cluster i in `stations.csv` (fixed for the term) and $a_{s,d}$ is the number of cleaned trips that *end* at station s on Oslo day d. The trip files are in UTC and are converted with the Europe/Oslo time zone, never a fixed offset. A closed station contributes 0. The forecast $\hat{y}_{i,d} \geq 0$ is a real number.

Origin and what is known there. Day d is forecast at the origin $t_0(d) = 21{:}00$ Oslo time on day $d-1$, so the horizon runs from 3 to 27 hours ahead. At $t_0(d)$ we may use:

- complete daily counts $y_{i,t}$ for $t \leq d-2$ (`cluster_daily`);
- the *partial* count of day $d-1$, trips that ended before 21:00, written $e_{i,d-1}$ (`cluster_before21`); in October–December this is about 92% of a day;
- the calendar of d (weekday, public holidays), known in advance;
- weather *forecasts* for d issued before $t_0(d)$ (`source="forecast"`); measured weather only for whole days up to about $d-3$.

We may *not* use the whole-day count of $d-1$, trips that end after 21:00 on $d-1$, or the actual weather of d: each of these is leakage. Training rows and the prediction row are built by the same code with the same as-of cut.

Success criterion. Over the five clusters and the D target days of a round,

$$\text{WMAPE} = \frac{\sum_{i=1}^{5}\sum_{d=1}^{D}|y_{i,d} - \hat{y}_{i,d}|}{\sum_{i=1}^{5}\sum_{d=1}^{D} y_{i,d}}, \qquad \text{bias} = \frac{\sum_{i,d}\hat{y}_{i,d}}{\sum_{i,d} y_{i,d}}, \qquad \text{skill} = 1 - \frac{\text{WMAPE}}{\text{WMAPE}_{B1}}.$$

A model is useful to us only if it beats the yardstick B1 (skill > 0) by more than the year-to-year noise, with a bias near 1.00. WMAPE weights each cluster by its volume, so the Center (about 54% of the inflow) dominates the score. The clusters' errors are strongly correlated (about 0.85), so a round of 9–12 days is worth little more than 9–12 independent forecasts; we treat differences of about 0.02 in WMAPE as noise.

Assumptions we check. (i) *Level shifts*: 2019 was about twice as busy as 2023–2025 and 2026 is about 8% above 2025, so old years may hurt rather than help. (ii) *Seasonality*: the rounds fall in the steepest part of the autumn decline, and the outer clusters fall faster than the Center. (iii) *Weekday structure* and (iv) *weather* (24 mm of rain on 8 October 2026 halved the city inflow). (v) *Stations* open and close, which moves a cluster's level slowly. (vi) *Clock*: daylight-saving days have 23 or 25 hours; all test days are ordinary.

2 Validation plan

Rolling origin, replaying the harness. For every validation day d we rebuild the information set as it stands at $t_0(d)$ (full days up to $d-2$, the before-21:00 count of $d-1$) and produce one forecast. Anything that is estimated (shares, factors, model coefficients) uses only data that was available at that origin, as `fit_model` does in the real run.

Folds: the test calendar in past years. The rounds are 9–20 November (R1, 12 days) and 23 November–1 December (R2, 9 days). We replay these two windows in 2022, 2023, 2024 and 2025: eight folds, 84 target days, 420 cluster-days. We do not validate on 2019–2021 (a level twice as high in 2019, the closure from 9 December 2019 to 17 March 2020, winter shutdowns before 2021), but a model may still use those years for training.

Three-way split in time. Features, settings and hyper-parameters are chosen on the 2022–2024 folds (*tune*). The 2025 folds, closest to the current regime, are the *check*: each candidate is scored there once, after its settings are fixed. We report every fold, because one year's November can be easy and the next one hard.

Sanity checks. (a) On 8 October–16 December of 2022–2025 our code reproduces the brief's yardsticks: WMAPE 0.252 for last week's value and 0.204 for B1 (bias 1.117), and all five per-cluster values to two decimals. This checks our data, time zones and the implementation of B1. (b) We also score 8 October–5 November 2026, the only weeks of the current, higher level. (c) The submitted file is run through the self-check kit on several cut dates (`selfcheck.py -cut-at`) to verify the two-function contract, the 15-minute and 6 GB limits, and that the weather it requests is present.

3 First baseline and its validation score

Yardstick B1. With complete days up to $d-2$ only,

$$\hat{y}^{B1}_{i,d} \;=\; \underbrace{\tfrac{1}{4}\big(y_{i,d-7}+y_{i,d-14}+y_{i,d-21}+y_{i,d-28}\big)}_{F_{i,d}\text{: same weekday, last four weeks}} \times \underbrace{\frac{\frac{1}{7}\sum_{k=2}^{8} y_{i,d-k}}{\frac{1}{28}\sum_{k=2}^{29} y_{i,d-k}}}_{r_{i,d}\text{: recent level}},$$

with $r_{i,d}=1$ when the 28-day mean is zero. The naive alternative is last week's value, $\hat{y}_{i,d}=y_{i,d-7}$.

First improvement: B1-evening × seasonal factor. Two changes, both using only what is known at the origin. *(1) Use the evening of $d-1$.* The share of a day that arrives before 21:00 is estimated from the last eight complete days with the same weekday as $d-1$, $\hat{p}_{i,d-1}=\frac{1}{8}\sum_{k=1}^{8} e_{i,d-1-7k}/y_{i,d-1-7k}$, and the whole of $d-1$ is estimated as $\tilde{y}_{i,d-1}=e_{i,d-1}/\hat{p}_{i,d-1}$. The level ratio then includes that day:

$$\tilde{r}_{i,d}=\frac{\frac{1}{7}\big(\tilde{y}_{i,d-1}+\sum_{k=2}^{7} y_{i,d-k}\big)}{\frac{1}{28}\big(\tilde{y}_{i,d-1}+\sum_{k=2}^{28} y_{i,d-k}\big)}, \qquad \hat{y}^{E}_{i,d}=F_{i,d}\,\tilde{r}_{i,d}.$$

(2) Correct the seasonal over-forecast. A trailing level ratio lags behind the November decline, so B1 forecasts too many bikes (Figure 1b). For a day d in year Y we take the realised ratios $y_{i,t}/\hat{y}^{E}_{i,t}$ of *earlier* years only, on days t within $\pm h$ days of the same calendar date (days on which the whole system was closed are left out), and use their median:

$$c_i(d)=\operatorname{median}\Big\{\, y_{i,t}/\hat{y}^{E}_{i,t} \;:\; t\in \text{years } 2019,\dots,Y-1,\; |t-d^{(t)}|\le h \Big\}, \qquad \hat{y}_{i,d}=c_i(d)\,\hat{y}^{E}_{i,d},$$

where $d^{(t)}$ is the date d moved to the year of t. The half-width $h=14$ days was chosen on the tune folds from $\{3,5,7,10,14,21,28\}$; the 2025 folds were scored once afterwards.

Table 1: WMAPE and bias (forecast/actual) of each model; skill against B1 over all 84 days. Tune = 2022–2024 folds (36 R1 days, 27 R2 days); check = 2025 folds.

	R1 tune		R2 tune		R1 2025		R2 2025		All 84 days		
Model	WMAPE	bias	WMAPE	bias	WMAPE	bias	WMAPE	bias	WMAPE	bias	skill
Last week's value	0.208	1.05	0.419	1.33	0.255	1.20	0.321	1.26	0.272	1.15	−0.29
B1 (yardstick)	0.166	1.08	0.305	1.27	0.201	1.19	0.272	1.23	0.211	1.15	0
B1 × seasonal factor	0.140	0.95	0.222	1.04	0.122	1.05	0.204	1.00	0.160	0.99	+0.24
B1-evening × seasonal factor	0.131	0.95	0.208	1.01	0.107	1.05	0.211	0.99	0.150	0.98	+0.29

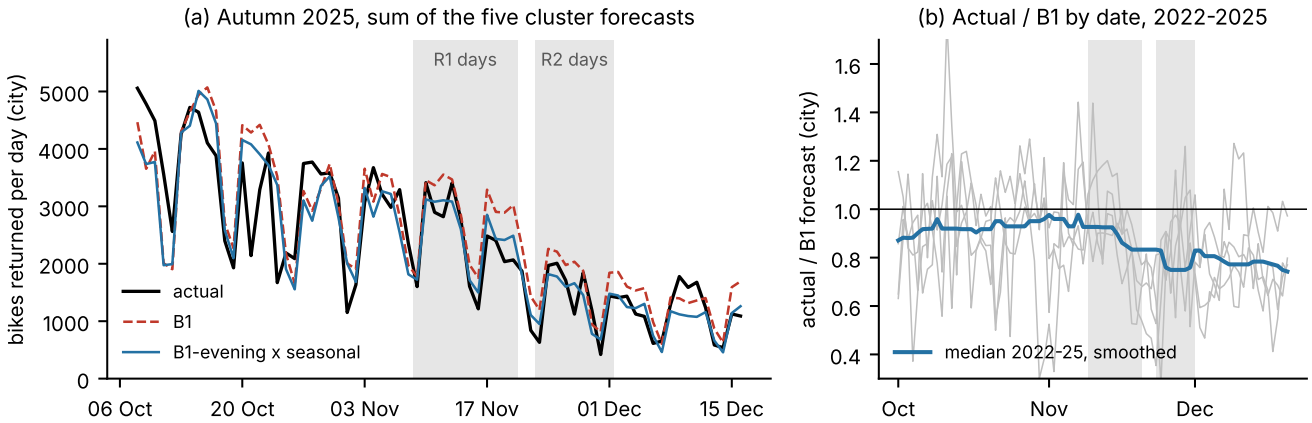


Figure 1: (a) City inflow in autumn 2025 (sum over the five clusters) with the B1 forecast and our first model; gray bands are the round windows replayed in 2025. (b) Ratio of actual inflow to the B1 forecast for each year 2022–2025 (gray) and its smoothed median (blue): below 1 through November, deepest in the R2 window.

What the numbers support. B1's error in November is largely systematic: its bias is 1.08 on the R1 tune days and 1.27 on the R2 tune days. The seasonal factor removes most of it (bias 0.95–1.05) and lowers WMAPE over the 84 days from 0.211 to 0.160; adding the evening of $d-1$ lowers it to 0.150 (skill +0.29). Per cluster, the North-west stays the hardest (WMAPE 0.194 against 0.289 for B1) and the Center the easiest (0.133 against 0.185). The spread across years is large: the eight folds range from 0.06 to 0.26 for our model and from 0.13 to 0.38 for B1, so the gain from the evening term (0.010) is within the noise band and must keep showing up before we rely on it. On 8 October–5 November 2026 the gain is smaller (B1 0.247, our model 0.231): in October the decline is mild and the day-to-day swings come from the weather, which neither model sees.

4 Next steps

1. **Freeze 1 (8 November).** Submit B1-evening × seasonal factor as `main.py`: `fit_model` estimates $c_i(\cdot)$ and $\hat{p}$ from the snapshot, `make_forecast` applies them, with a fall-back to B1 if a value is missing. Checked with the self-check kit before every push.
2. **Weather.** Request hourly precipitation and temperature *forecasts* at the five cluster centres (archive from January 2025), aggregate them to Oslo-day features (daytime rain, mean temperature), and model the remaining log-ratio $\log(y_{i,d}/\hat{y}_{i,d})$. We train on forecasts, not on measured weather, so that the training rows match the prediction row.
3. **A learned model.** A regularised regression and a gradient-boosted tree on the log scale (lags, level ratios, weekday, day of year, weather), with an absolute-error loss that matches WMAPE; one model per cluster against one pooled model, and a bottom-up sum of station forecasts. Each is compared on the same folds and kept only if it beats the current model on the tune folds by more than the noise band.
4. **The 2026 regime.** Check whether factors learned from 2019–2025 still hold at the higher 2026 level, especially for the fast-growing North and North-west clusters.

AI Use Note

Tools. Claude (Anthropic), used in the Claude desktop app with read access to the course files, the project brief and the self-check kit. **What we asked.** To summarise the brief's and the kit's rules; to write the backtest code (B1, last week's value, the seasonal and evening variants on the replayed windows) and the figure; to draft the text of this checkpoint. **What we changed.** [2–3 sentences chosen from the box below]

TASLAK – grup için öneriler. Teslimden önce bu kutuyu silin.

Yalnızca **gerçekten yaptığımız** şeyleri seçip düzenleyelim; 2–3 cümle yeter. Hocanın en çok değer verdiği not somut ve kontrol edilebilir olanıdır: neyi kontrol ettik, neyi yanlış ya da eksik bulduk, neyi neden değiştirdik. "Metni okuduk" gibi genel bir cümle tek başına zayıf kalır. Her seçeneğin yanında, onu doğru kılmak için yapılması gereken iş yazıyor.

(A) Kendi cümlelerimiz. *Yapılacak: 1. bölümü brief'in 1.2 ve 2.2 bölümleriyle karşılaştırıp kendi cümlelerinizle yeniden yazın.*
"We rewrote Section 1 in our own words after checking each item of the information set against Sections 1.2 and 2.2 of the brief, and we added [. . .] ourselves."

(B) Elle doğrulama. *Yapılacak: durum özetindeki alıştırmayı hesap makinesiyle yapın.*
"We recomputed B1 and our forecast by hand for the Center cluster on 13 November 2025 (1926.4 and 1753.3 against an actual 1916) and matched the code. The same example showed us that a model that is better over 84 days can still lose to B1 on a single day."

(C) Kendi sorduğumuz bir test. *Yapılacak: bir varsayıma itiraz edin. Testi ben çalıştırırım, kararı siz verirsiniz.*
"We questioned whether 2019, a year twice as busy and closed from 9 December, should enter the seasonal factor. Dropping it changed the tune-fold WMAPE from [x] to [y], which is within the noise band, so we [kept / dropped] it."

(D) Fazla iddialı bir ifadeyi düzeltmek. *Yapılacak: okurken abartılı bulduğunuz bir cümleyi işaretleyin.*
"We changed '[. . .]' to '[. . .]' because the 84 validation days do not support the stronger claim."

(E) Bir öneriyi reddetmek. *Yapılacak: AI'nın önerdiği bir şeyi gerekçeyle reddettiyseniz yazın.*
"We did not adopt the assistant's suggestion to [. . .] because [. . .]."

What we verified. Our B1 and last-week scores reproduce the brief's yardstick (0.204 and 0.252, bias 1.12, all per-cluster values to two decimals); summing `station_day_arrivals` over each cluster's stations gives `cluster_daily_inflow` exactly; every forecast for day d is built from data available at 21:00 on $d-1$ (shifts checked in the code); and h was chosen on the 2022–2024 folds before the 2025 folds were scored. Our `main.py`, run through the self-check kit at three past origins, gives exactly the forecasts of the backtest.